Algorithms for DNA Sequencing

Today's goal • 3/3 items

Module 1
DNA sequencing, strings and matching

Module 2
Preprocessing, indexing and approximate matching

Module 3
Edit distance, assembly, overlaps

Module 3: Edit distance, assembly, overlaps

- Module 3 Introduction
  Video • 1 min
- Lecture: Solving the edit distance problem
  Video • 12 min
- Lecture: Using dynamic programming for edit distance
  Video • 12 min
- Practical: Implementing dynamic programming for edit distance
  Video • 6 min
- Lecture: A new solution to approximate matching
  Video • 9 min
- Lecture: Meet the family: global and local alignment
  Video • 10 min
- Practical: Implementing global alignment
  Video • 8 min
- Lecture: Read alignment in the field
  Video • 4 min
- Lecture: Assembly: working from scratch
  Video • 2 min
- Lecture: First and second laws of assembly
  Video • 8 min
- Lecture: Overlap graphs
  Video • 8 min
- Practical: Overlaps between pairs of reads
  Video • 4 min
- Practical: Finding and representing all overlaps
  Video • 3 min

Quiz

- Module 3
  Graded Assignment • Grade: 100%

Programming Homework

- **Programming Homework 3 Instructions (Read First)**
  Reading • 10 min
- Programming Homework 3
  Graded Assignment • 30 min

Module 4
Algorithms for assembly

Module 4: Algorithms for assembly

- Module 4 introduction
  Video • 1 min
- Lecture: The shortest common superstring problem
  Video • 8 min
- Practical: Implementing shortest common superstring
  Video • 4 min
- Lecture: Greedy shortest common superstring
  Video • 7 min
- Practical: Implementing greedy shortest common superstring
  Video • 7 min
- Lecture: Third law of assembly: repeats are bad
  Video • 5 min
- Lecture: De Bruijn graphs and Eulerian walks
  Video • 8 min
- Practical: Building a De Bruijn graph
  Video • 4 min
- Lecture: When Eulerian walks go wrong
  Video • 9 min
- Lecture: Assemblers in practice
  Video • 8 min
- Lecture: The future is long?
  Video • 9 min
- Lecture: Computer science and life science
  Video • 5 min
- Lecture: Thank yous
  Video • 43 sec

Homework

- Programming Homework 4
  Graded Assignment • 30 min

# Programming Homework 3 Instructions (Read First)

Practice with Lab Sandbox

0:08 / 1:20 2x

We saw how to adapt dynamic programming to find approximate occurrences of a pattern in a text. Recall that:

- Rows of the dynamic programming matrix are labeled with bases from P and columns with bases from T
- Elements in the **first row are set to 0**
- Elements in the first column are set to 0, 1, 2, ..., as for edit distance
- Other elements are set in the same way as elements of a standard edit distance matrix
- The **minimal value in the bottom row** is the edit distance of the closest match between P and T

First, download the provided excerpt of human chromosome 1

https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/chr1.GRCh38.excerpt.fasta

Second, parse it using the readGenome function we wrote before.

Third, adapt the editDistance function we saw in practical (copied below) to answer questions 1 and 2 below. Your function should take arguments p (pattern), t (text) and should return the edit distance of the match between P and T with the fewest edits.

```
def editDistance(x, y):
    # Create distance matrix
    D = []
    for i in range(len(x)+1):
        D.append([0]*(len(y)+1))
    # Initialize first row and column of matrix
    for i in range(len(x)+1):
        D[i][0] = i
    for i in range(len(y)+1):
        D[0][i] = i
    # Fill in the rest of the matrix
    for i in range(1, len(x)+1):
        for j in range(1, len(y)+1):
            distHor = D[i][j-1] + 1
            distVer = D[i-1][j] + 1
            if x[i-1] == y[j-1]:
                distDiag = D[i-1][j-1]
            else:
                distDiag = D[i-1][j-1] + 1
            D[i][j] = min(distHor, distVer, distDiag)
    # Edit distance is the value in the bottom right corner of the matrix
    return D[-1][-1]

```

Hint: In the "A new solution to approximate matching" video we saw that the best approximate match of P =GCGTATGCwithin T =TATTGGCTATACGGTThad 2 edits. You can use this and other small examples to double-check that your function is working.

Go to next item ✓ Completed

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